

Code-Layout, Readability and Reusability

Date	1 July 2026
Project Name	OptiCrop
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the OptiCrop codebase — the Smart Agricultural Production Optimization Engine — in terms of its structure, readability, and reusability. OptiCrop integrates environmental factors such as Nitrogen (N), Phosphorous (P), Potassium (K), soil temperature, humidity, pH, and rainfall to recommend suitable crops and assess crop-environment suitability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development, including by agricultural researchers, policymakers, and stakeholders.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation is maintained across Python, HTML, CSS, and JavaScript files.
2	Proper File Structure	Files and folders are logically organized (data, models, notebooks, app, utils)	Yes	Project follows a structured layout with separate folders for model, templates, static files, and datasets.
3	Meaningful Variable Names	Variables reflect their purpose clearly (e.g., N, P, K, temperature, humidity, ph, rainfall)	Yes	Variables use descriptive names such as N, P, K, temperature, humidity, ph, and rainfall.
4	Function / Method Names	Functions are descriptively named (e.g., predict_crop(), preprocess_input(), load_model())	Yes	Functions are named according to their functionality,

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				improving readability and maintainability.
5	Code Comments	Inline and block comments explain logic, especially data preprocessing and model prediction steps	Partial	Comments are included in important sections, though additional comments could further improve clarity.
6	Modular Design	Code is split into reusable functions/modules (data handling, model training, prediction, UI)	Yes	The project separates machine learning, preprocessing, prediction logic, and frontend components into reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unnecessary and duplicate code has been minimized to improve maintainability.
8	Error Handling	Exceptions and errors (invalid input, missing values) are handled gracefully	Partial	Basic input validation is implemented; advanced exception handling can be added in future versions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module (scaling, encoding N-P-K/climate inputs)	Python (Pandas, NumPy, Scikit-learn)	Scenario 1, 2 & 3 – input validation and preparation	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
2	Crop Recommendation Model (predict_crop function)	Python (Scikit-learn / ML model)	Scenario 1 – Smart Crop Recommendation for Farmers	High
3	Crop Suitability & Environmental Assessment Module	Python (Flask/Django backend)	Scenario 2 – Crop Suitability and Environmental Assessment	High
4	Data Analytics & Visualization Module	Python (Pandas, Matplotlib/Seaborn)	Scenario 3 – Agricultural Research and Policy Planning	High
5	Input Form / UI Component	HTML, CSS, JavaScript (Web App)	Scenario 1 & 2 – farmer and user data entry	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend, backend, and machine learning components.
Readability	5	Meaningful variable names, consistent formatting, and clear code flow improve readability.
Reusability	5	Prediction logic, preprocessing, and UI components are reusable across different modules.
Documentation / Comments	4	Important sections are documented; additional comments could further improve maintainability.
Overall Score	4.75/5	The project demonstrates good coding practices, modularity, readability, and effective integration of machine learning with the web application.